Prerna Arote

Security & Privacy Researcher | Digital Financial Systems | Applied Cryptography (ZKP, MPC)



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Profile

I am a Postdoctoral Researcher in the Distributed Systems and Networks (DistSys) group at IMDEA Networks, Madrid, Spain. I received my Ph.D. from the Indian Institute of Science (IISc), Bangalore, where I specialized in blockchain security.
My research lies at the intersection of secure distributed systems, blockchain security, and decentralized finance (DeFi) security. I design secure and scalable protocols for decentralized systems, with interests spanning blockchain protocols, payment channels, privacy-preserving technologies, and empirical security analysis. My current work focuses on private trading systems and the security of decentralized financial infrastructure.

Education

Aug 2017 – July 2025
(CGPA: 8.66/10)

Ph.D., Department of Electronic Systems Engineering, Indian Institute of Science, Bangalore.
Advisor: Prof. Joy Kuri
Thesis title: *Enhancing Blockchain Security and Efficiency: Solutions for Micropayments, Payment Channels, and IoT Applications.*
Thesis Summary: We studied double-spending attacks (DSA) in Bitcoin, analyzing their root causes and preventive measures. To mitigate DSA in micropayments, reducing confirmation latency is essential. Existing off-chain payment channel network (PCN) solutions address latency but are vulnerable to attacks like Fakey, Griefing, and Wormhole. We proposed cryptographic commitment-based solutions to detect and prevent these attacks and developed a method to reduce confirmation latency without locking customer collateral, unlike traditional payment channels. Additionally, we applied privacy-preserving techniques using the Ethereum blockchain to Industrial IoT (IIoT)-based applications, including calibration hierarchies and multi-access edge computing.

Aug 2011 – Aug 2013
(CGPA: 8.76/10)

M.Tech. Computer Science specialization Information Security, IIIT Gwalior, M.P.
Advisor: Prof. Karm Veer Arya
Thesis title: *Detection and Prevention of ARP Spoofing Attack using Centralized Approach.*
Thesis Summary: Developed an ARP poisoning detection algorithm using the ICMP protocol and a prevention algorithm utilizing the ARPSpoof tool. Both algorithms were implemented in a private network consisting of four systems running the BackTrack OS.

Research Interests

Secure Distributed Systems, Blockchain Security and Scalability , Decentralized Finance, and Privacy-Preserving Protocols.

Research Projects

Privacy Gaps in MPC-Based Decentralized Dark Pools (PI, Postdoc)
This project evaluates the decentralized dark pool Renegade and finds that its cryptographic design does not achieve true dark pool guarantees in practice. While MPC and zero-knowledge proofs ensure correctness, they fail to provide fairness, enabling selective aborts that leak sensitive trading information. The system's discovery layer and probing attacks further compromise both pre- and post-trade privacy. Additionally, centralization in relayers and protocol-level weaknesses introduce liveness and security risks, effectively recreating the very information asymmetry dark pools aim to eliminate.

Securing HTLCs in Lightning Networks (PI, Ph.D. project under supervision of Prof. Joy Kuri)
Existing off-chain payment channel solutions suffer from various attacks such as Fakey, Griefing, and Wormhole attacks. In these attacks, adversaries aim to disrupt transaction throughput by exhausting channel capacity or stealing routing fees from honest nodes along the payment path. We propose a scheme called "CommTLC," which leverages Pedersen commitments and signatures to detect and punish adversaries in Fakey, Griefing, and Wormhole attacks. We implement the proposed scheme and analyze its security within the Universal Composability (UC) framework. Additionally, we compare the performance of CommTLC with the latest schemes, MAPPCN-OR and EAMHL+.

Latency Reduction in Bitcoin Micropayments (PI, Ph.D. project under supervision of Prof. Joy Kuri)
In Bitcoin micropayments, the challenge of reducing confirmation latency without locking users' funds remains unresolved. Therefore, we propose an on-chain solution "Q_pay," to expedite the validation of Bitcoin payments without relying on users' collateral. The demonstrated protocol guarantees that sellers are guaranteed payment and can provide fast service to buyers once a "committee" approves the Q_pay transaction. Certain committee members with mining capabilities are responsible for mining a Q_block, and the time required to mine a Q_block is significantly shorter than that of a regular Bitcoin block due to its lower difficulty target. We evaluate Q_pay using 600 Bitcoin nodes on an emulated network.

Privacy-Preserving and Traceable Calibration Hierarchy (PI, Ph.D. project under supervision of Prof. Joy Kuri)
Metrological traceability requires linking measurements to trusted standards, but doing so can expose sensitive identities and relationships. We propose DTH, a decentralized system for secure certificate storage, verification, and revocation. Using Merkle trees and zkSNARKs, DTH preserves privacy by hiding certificate data, holder identities, and hierarchical links.

Secure and Reliable Task Offloading Framework (PI, Ph.D. project under supervision of Prof. Joy Kuri)
IoT task offloading via intermediary devices exposes sensitive data and risks collusion in untrusted environments. We propose SC-TOF, a smart contract–based scheme using multi-signatures to securely discover edge servers without intermediaries, preserving privacy. Implemented with iFogSim and Ethereum, it completes 20 MB and 200 MB tasks in 0.8 s and 5.2 s, respectively.

Secure calibration in high-assurance IoT: traceability for safety resilience (co-PI in a project collaboration with University of Strathclyde.)
We proposed a novel resilient architecture for calibration infrastructure, implemented using smart contracts on the Ethereum blockchain, and evaluated its functionality and scalability.

Research Publications

Journal papers

[1] **Prerna Arote** and Joy Kuri. "Q_Pay: A Low-Latency Bitcoin Micropayment Protocol for Instant Service Access," **SN Computer Science Journal**," volume 7.
[2] **Prerna Arote** and Joy Kuri. "CommTLC: An Alternative to Reduce the Attack Surface of HTLCs in Lightning Networks." **IET Blockchain** 5(1), 2025: e70005.

Conference papers

[3] **Prerna Arote**. "Mind the Gap: Liquidity Poisoning Attacks on Multihop Cross-Chain Arbitrage," [Paper] at IEEE International Conference on Blockchain and Cryptocurrency **(ICBC)**, Brisbane, Australia, 2026.
[4] **Prerna Arote**. "Breaking Sandwich MEV in ERC-20 DeFi via Context-Aware Directional Constraints," [Poster] at IEEE International Conference on Blockchain and Cryptocurrency **(ICBC)**, Brisbane, Australia, 2026.
[5] **Prerna Arote** and Joy Kuri. "SC-TOF: A Secure and Reliable Task Offloading Framework for IoT Devices Using Smart Contracts and Edge Computing," in 28th IEEE Conference on Innovation in Clouds, Internet and Networks **(ICIN)**, Paris, France, 2025.
[6] **Prerna Arote** and Joy Kuri. "Detect and Isolate an Adversary in Fakey and Griefing-R Attack on Lightning Network." In 2024 IEEE International Conference on Blockchain **(IEEE Blockchain)**, pp. 563-568. IEEE, Copenhagen, Denmark, 2024.
[7] **Prerna Arote** and Joy Kuri. "A Study of Double-spending Causes and Countermeasures in Bitcoin Network." In 2024 6th International Conference on Blockchain Computing and Applications **(BCCA)**, pp. 544-557. IEEE, Dubai, UAE, 2024.
[8] **Prerna Arote** and Joy Kuri. "ZCC: Mitigating Double-spending Attacks in Micropayment Bitcoin Transactions." In 2022 Fourth International Conference on Blockchain Computing and Applications **(BCCA)**, pp. 245-252. IEEE, Texas, USA, 2022.
[9] **Prerna Arote** and Joy Kuri. "Hybrid decentralized solution for bitcoin zero-confirmation transactions." In 2019 11th International Conference on Communication Systems and Networks **(COMSNETS)**, pp. 1-5. IEEE, Bangalore, India, 2019.
[10], Ryan Shah, Michael McIntee, Shishir Nagaraja, Sahil Bhandary, **Prerna Arote**, Joy Kuri, "Secure calibration in high-assurance IoT: traceability for safety resilience," 2019 (Arxiv).
[11] **Prerna Arote** and Karam Veer Arya. "Detection and prevention against ARP poisoning attack using modified ICMP and voting." In 2015 International conference on computational intelligence and networks **(CINE)**, pp. 136-141. IEEE, 2015.

In review.

[12] **Prerna Arote**, Adrian Saiz, Oriol Saguillo, Lucianna Kiffer. "Cryptographic Security Is Not Enough: Privacy Gaps in the Renegade Decentralized Dark Pool," arXiv pre-print is available at: https://arxiv.org/pdf/2609.27100.
[13] **Prerna Arote**, Adrian Saiz, Lucianna Kiffer. "SoK: Privacy-Preserving Order Matching," arXiv pre-print is available at: https://arxiv.org/pdf/2610.12535.

Work Experience

Dec 2025 - current	**Postdoc Researcher** IMDEA Networks, Madrid, Spain
Dec 2016 - July 2017	**Software Engineer** DIGITS, Bangalore
Mar 2015 - Mar 2016	**Project Associate** DIGITS, Bangalore
Sept 2013 - Nov 2014	**Assistance professor** in CSE department at Symbiosis Institute of Technology, Pune

Scientific Service

Topic Editor	Frontiers in Blockchain Journal (2025); Research Topic "Security, Economics, and MEV in Decentralized Finance Systems" (2025–26).
TPC Member	FC (2027), BCCA (2025)
Reviewer	**Journals:** Journal of Computer Security (JCS), SN Computer Science, Digital Communications and Networks (DCN), Computers & Security (C&S), Frontiers in Blockchain, Financial Innovation, Blockchain research and Applications Journal.
	Conferences: FC (2026), Blockchain (2025), COMSNETS (2022).
External Reviewer	IEEE S&P (2026), AFT (2026), ACM IMC (2026)
Organizations	IEEE ComSoc Member.

Technical Skills

Computer Languages	Solidity, Rust, C++, Python, Golang.
Software and Tools	Git, Docker, Kubernetes, ZKP, Ettercap, ARPSpoof, TcpDump, Wireshark, MySQL, LaTex, MS Office, Linux/Windows/kali OS.
Blockchain Platforms	Bitcoin, Ethereum, Solana, Arbitrum, Polygon, Sui.

Honors and Awards

2017-2023	Received **MHRD scholarship** during Ph.D. program.
2012-13	**Placement representative** for the academic year 2012-13 during post-graduation at IIITM.
2011-2013	Received **MHRD scholarship** during M.Tech program.
2011	**Secured All India Rank 5794** in Graduate Aptitude Test Engineering(GATE) for Computer Science and Information Technology conducted by MHRD out of 136027 students with 96 percentile.
2010	**Institute Award**, Awarded the Topper award three consecutive times for academic excellence that is ranked 1 among batches 2010 at PES College of Engineering.

References

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